Using

Semantic Link

To get an overview of Power BI in the lands of Middle Earth

SESSION TARGET

- Skilled python coder
- Expecting deep dive
- Best practices

- Never written python
- First steps into fabric monitoring

FORODWAITH
ERED MITHRIN
Withered Heath
IRON HILLS
RHÛN
Angmar
Carn Dûm
Here was of old the Witch-realm of Angmar
Icebay of Forochel
Himling
The Lost Realm of ARNOR
Ettenmoors
Erebor
The Lonely Mountain
Esgaroth
the Long Lake
Mountains of Mirkwood
Celduin (River Running)
Carnen (Redwater)
MIRKWOOD
Fornost
North Downs
Nenuial (Lake Evendim)
Weather Hills
Weathertop (Amon Sûl)
RHUDAUR
Trollshaws
Last Bridge
Bruinen (Loudwater)
Forlindon
Forlond
Mithlond (Grey Havens)
Tower Hills
THE SHIRE
Hobbiton
Brandywine Bridge
Old Forest
Bree
East-West Road
South Downs
CARDOLAN
Harlond
Harlindon
Blue Mountains
Baranduin (Brandywine)
Tharbad
MINHIRIATH
Eryn Vorn
East Bight
Sea of Rhûn
BROWN LANDS
N
The Realm of MIDDLE EARTH
Miles
50
100
150
200
Dagorlad (Battle Plain)
ERED LITHUI (Ash Mountains)
Udûn
Isenmouthe
Orodruin
Barad-dûr
Plateau of Gorgoroth
MORDOR
NURN
Sea of Nûrnen
EPHEL DUATH (Mountains of Shadow)
Osgiliath
Lebennin
Andrast (Ras Morthil)
Tolfolas
Ethir Anduin
South Gondor (now a debatable and desert land)

Hobbits

Small teams

Know each other

Tradition

No formal agreement

Innovation by individuals happens

Hobbits Organisational style

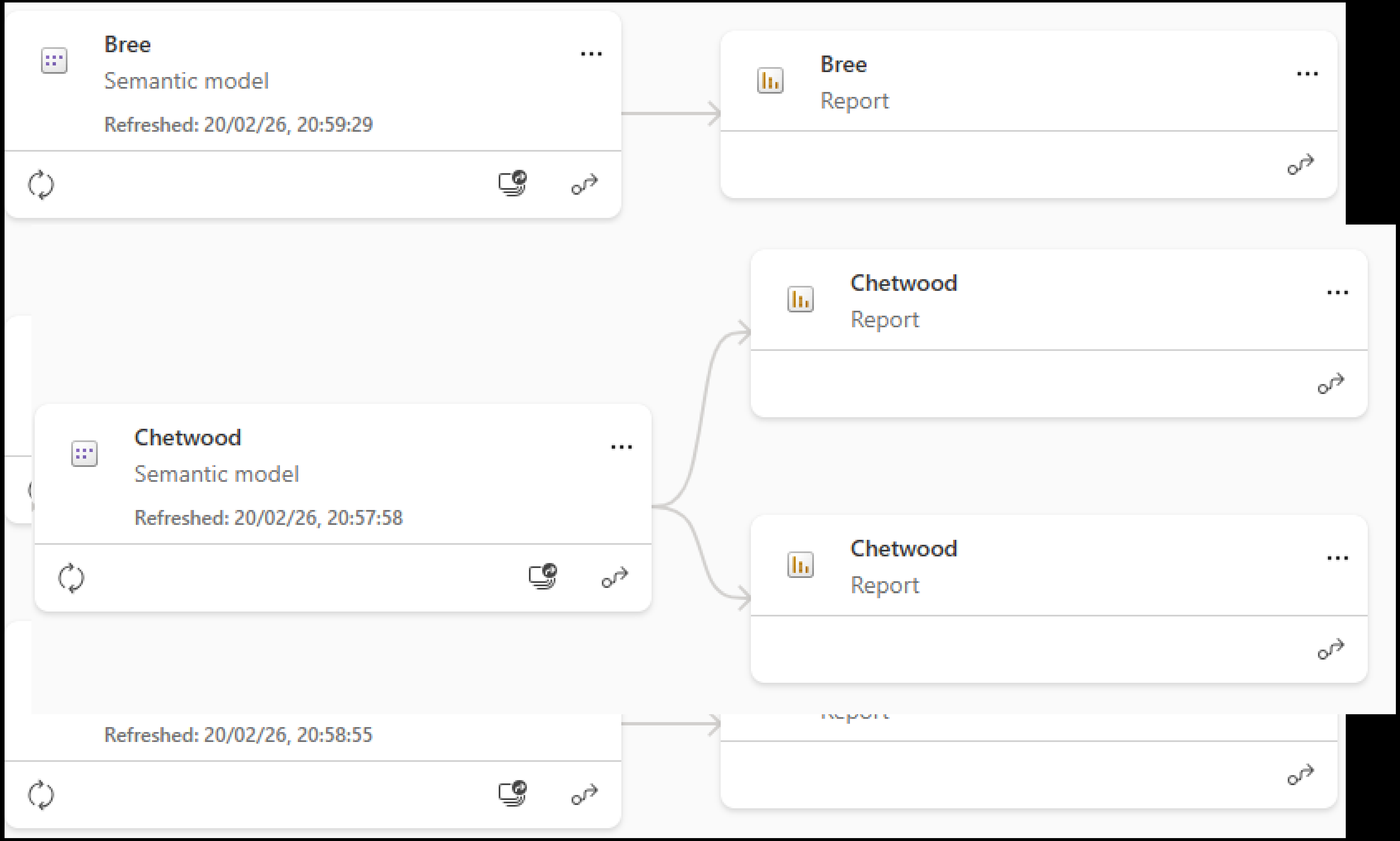
Hobbit workspaces

1 model => 1 report

Because…

resaons

Until someone tries something, for good or bad

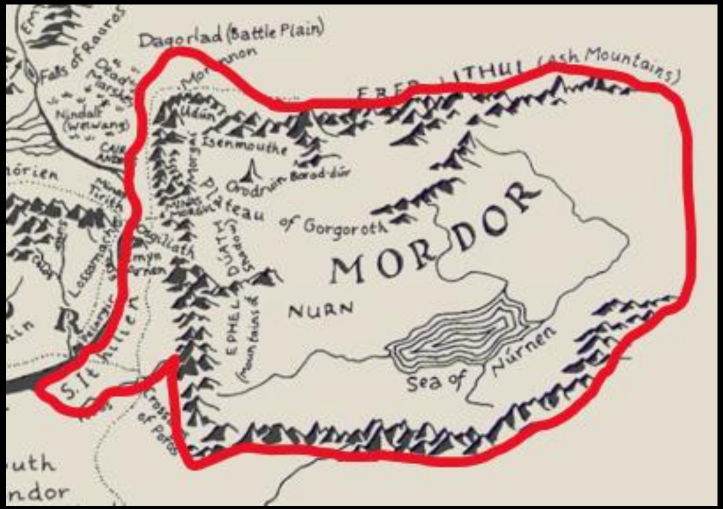


MORDOR

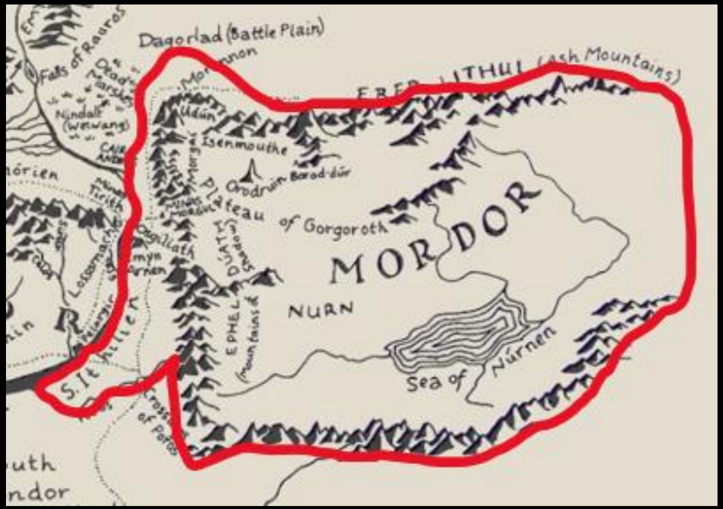
EXTREMELY STRONG ORGANISATION

UNIFORM LOOK AND FEEL

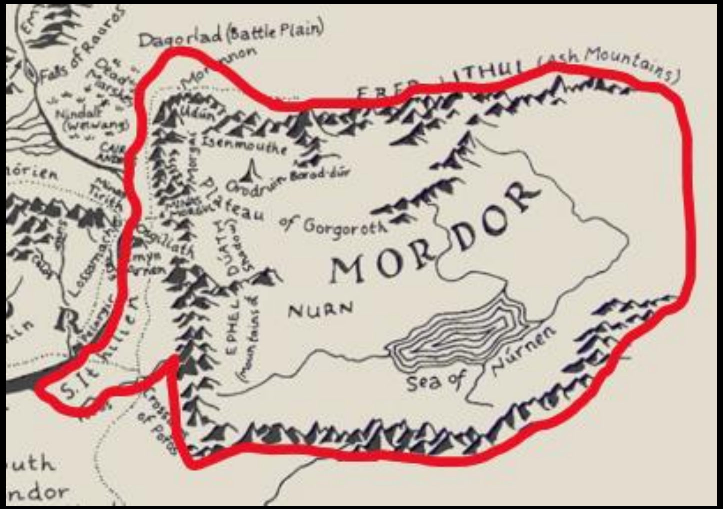
LOW INNOVATION

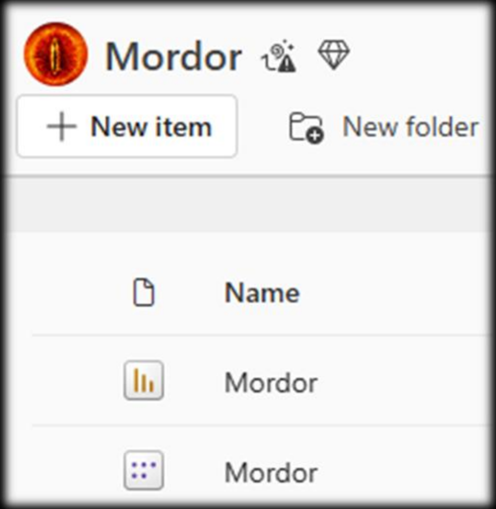


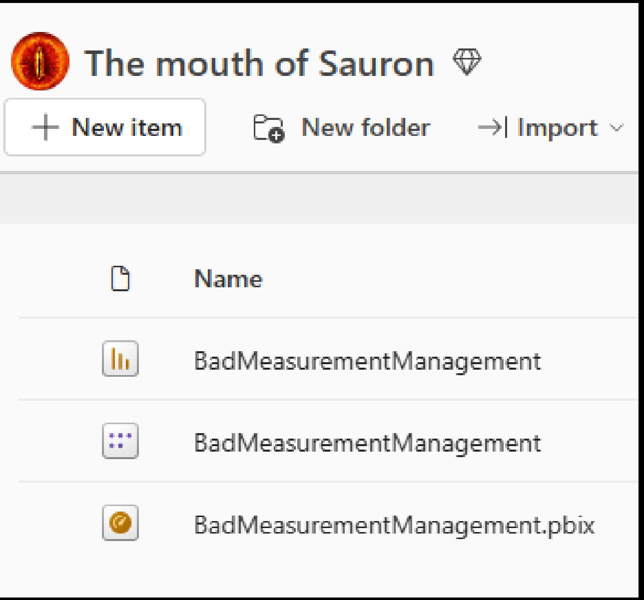
Mordor organisational style



Mordor Workspaces







Misty Mountains Goblins

No organisation

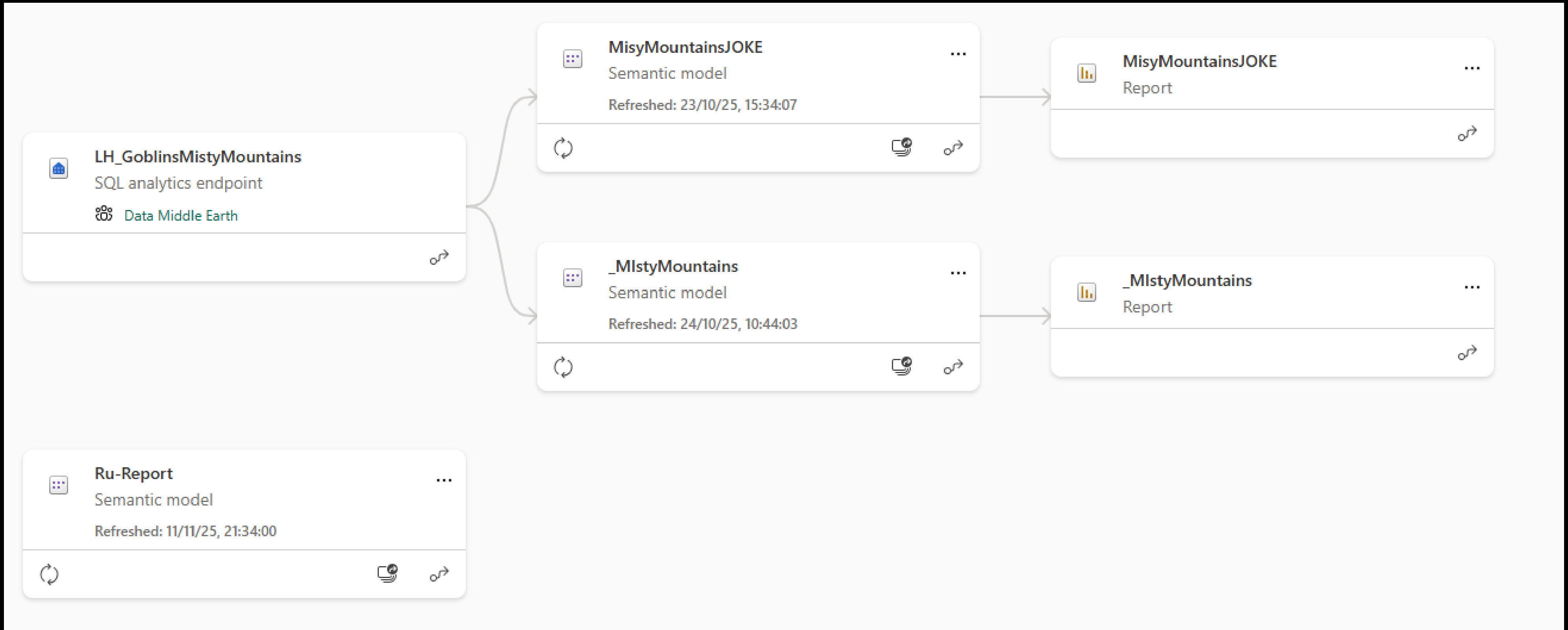
No agreements

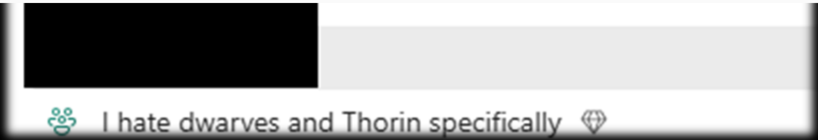
No uniformity

Free for all

Misty Mountains Goblins Organisational style

Misty Mountains Goblin workspaces





Isengard

Quick growing organisation

No time to organise

Chaos is the word

Isengard Organisational style

Overal organisation: work in progress

Loose

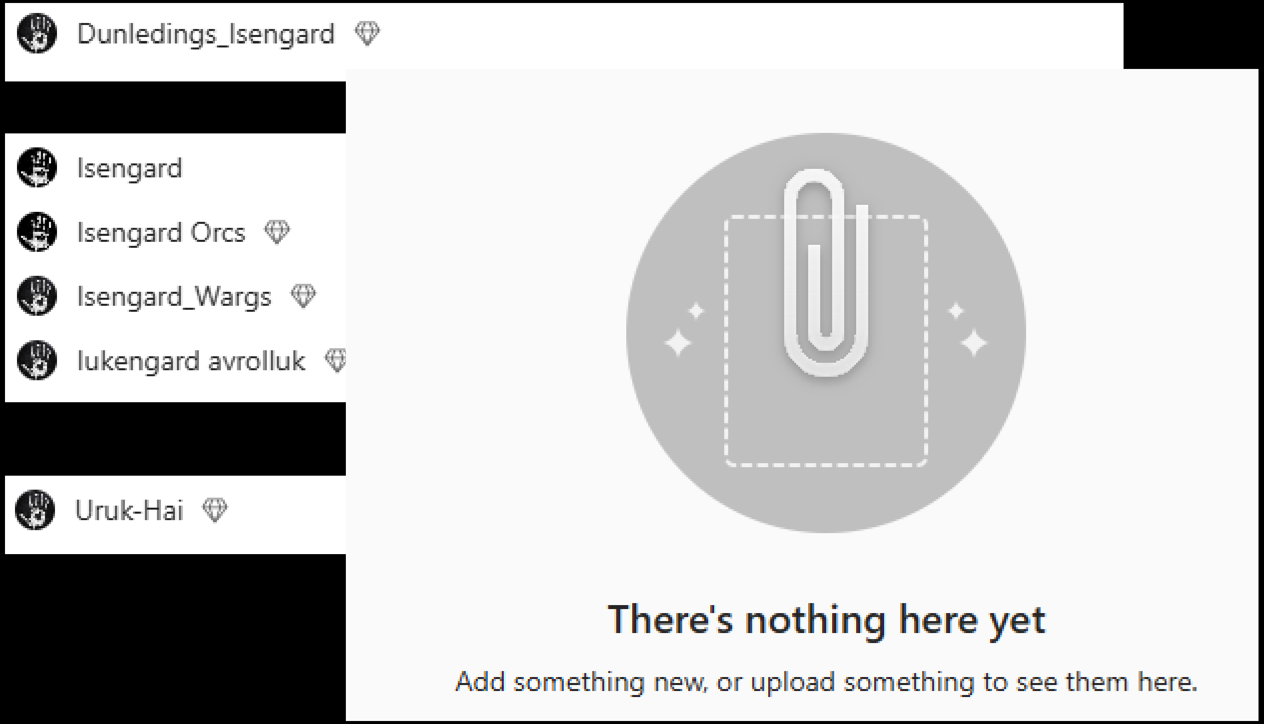
None

Strict

Isengard workspaces

Naming convention not yet agreed on, but the idea is there

Content?
We'll get to it

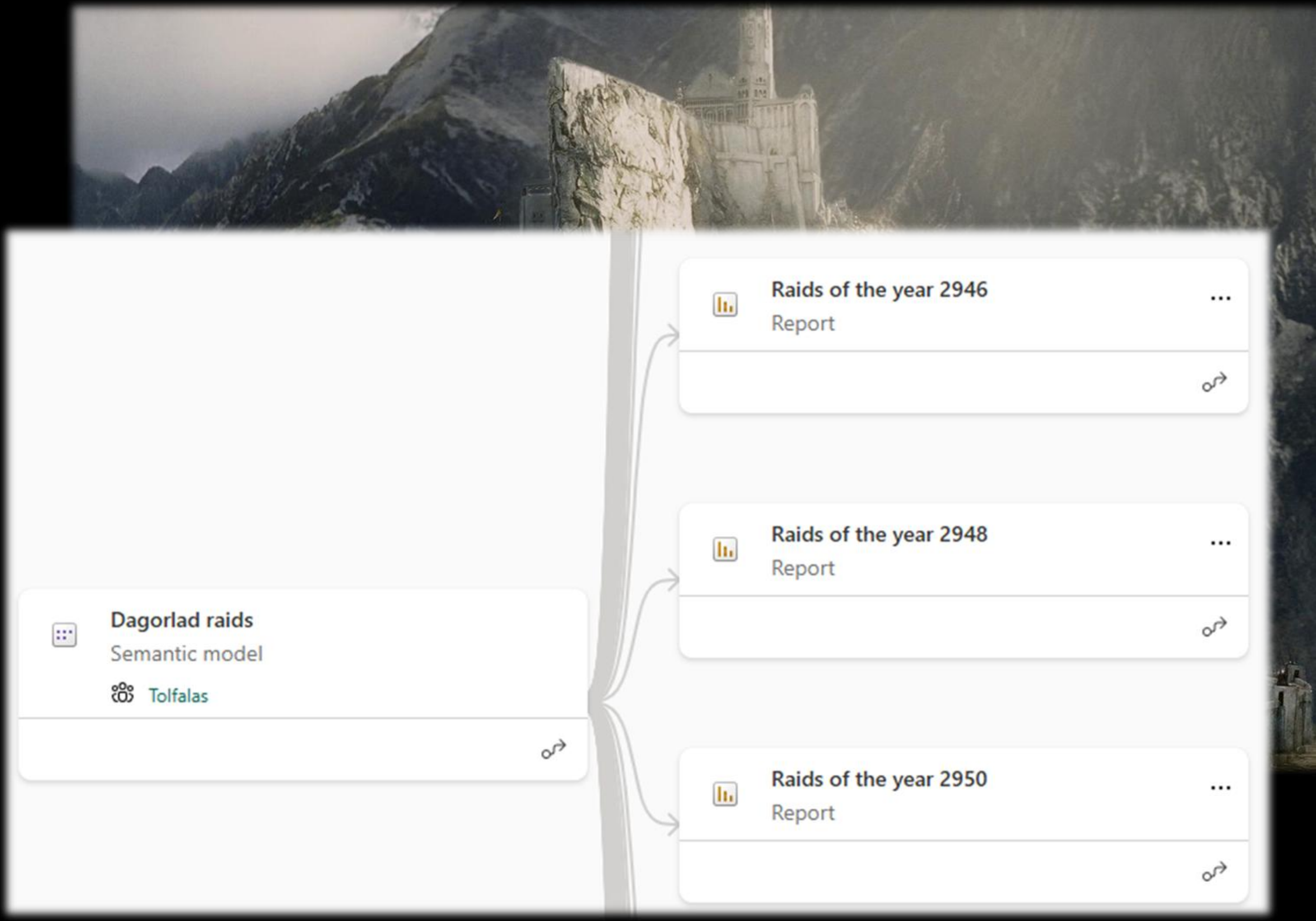


Gondor

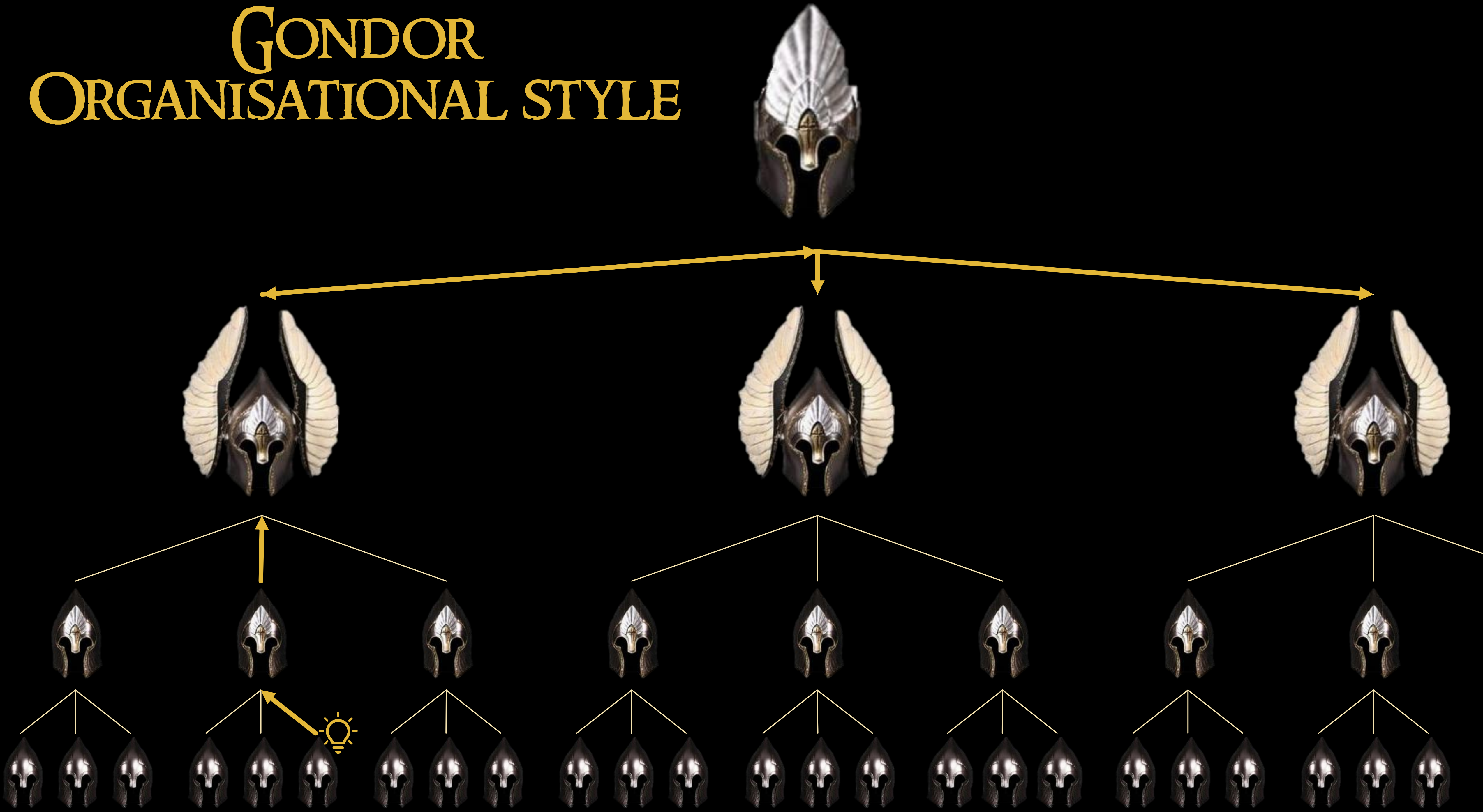
Strong organisation

Little to no differences

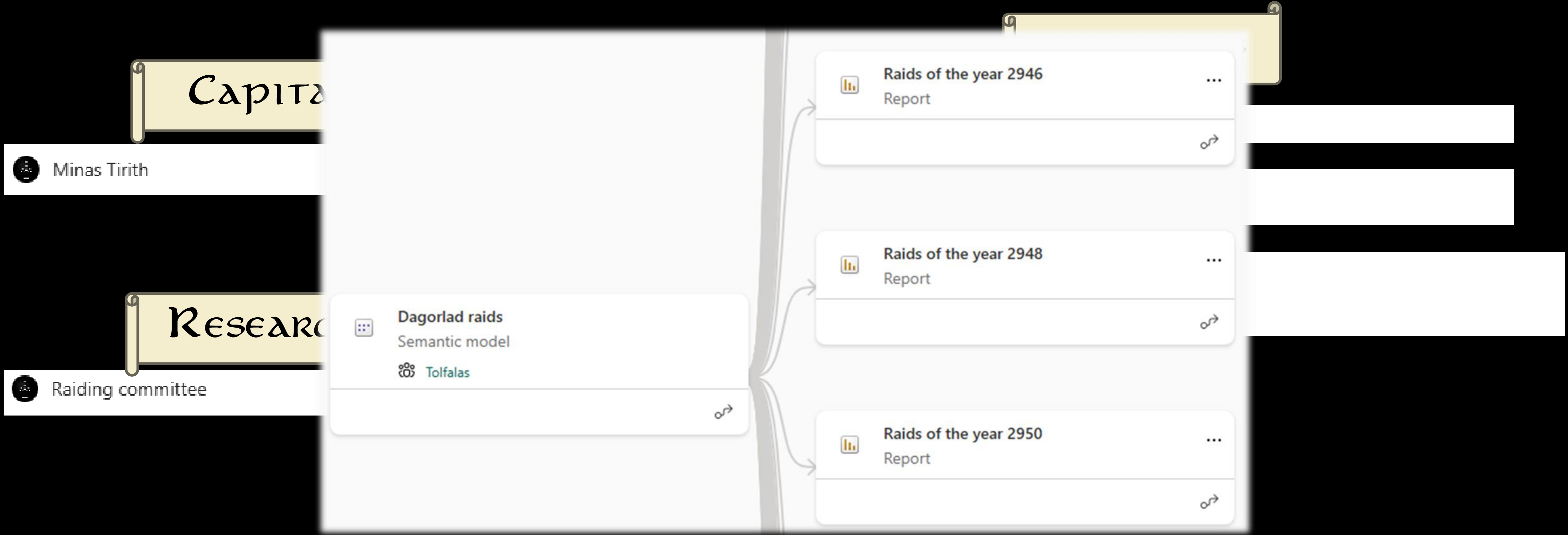
uniform look and feel



Gondor Organisational style



Gondor workspaces



Teams of Middle Earth

Centralised

Common theme

Many measures

Agreements

Departementalised

Decentralised

Innovation

Tradition

No measures

Bad relationships

Good semantic models

ADMIN

The yellow wizard

- How to get an overview
- What is fabric
- Python and notebooks
- What is Semantic link
- Without tenant powers

MIDDLE EARTH & US

Before Coffee

Story of the yellow wizard

Real-Life commentary by me

Jo David – My journey

C#
SSRS
TSQL

- Development
- Training
- Coaching
- Governance
- …

Sparkle

We are Sparkle

Your holistic data partner

- We integrate all relevant areas of expertise into a solution that brings value
- We combine in-depth knowledge about data (technology), with the skills to address business challenges
- We focus on long term partnerships with our clients and experts

People

50+ experts - Average +15 years of experience

Values and culture

Enthusiasm, Humility and Teamwork

Offices

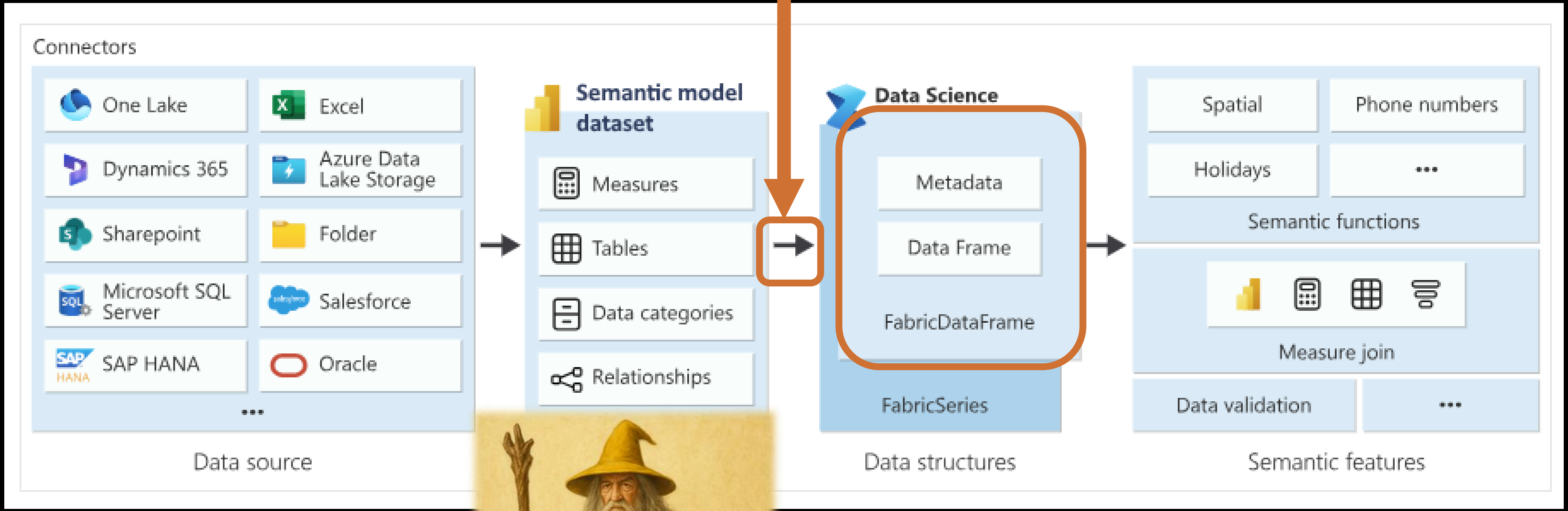
BENELUX - Leuven (HQ/B Louvain-La-Neuve (B), Hasselt (B), Antwerp (B)
Estonia – Tallinn

Part of the Cronos Group, 9000+ experts in Europe

What is semantic link

- Python SDK
- Exposes the Semantic model as Python objects
- Can read and modify model meta
- Used in Fabric notebooks

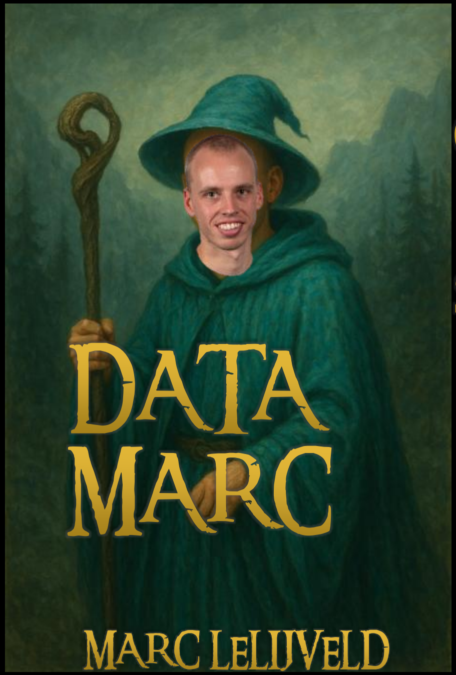
WHAT IS SEMANTIC LINK?



The bridge in the gap between Power BI & the synapse data science experience

FabricDataFrame is the data structure it uses

USAGE OF SEMANTIC LINK



- G SEMANTIC MODELS
- S

- GOVERNANCE
  - TENANT LEVEL
  - CAPACITY LEVEL
  - BUSINESS LEVEL

mssparkutils.session.stop()

Prerequisits

InvalidHttpRequestToLivy: [TooManyRequestsForCapacity] This spark job can't be run because you have hit a spark compute or API rate limit. To run this spark job, cancel an active Spark job through the Monitoring hub, choose a larger capacity SKU, or try again later. HTTP status code: 430 {Learn more} HTTP status code: 430.

- A Microsoft Fabric subscription
- Sign in to Microsoft Fabric
- Create a new notebook or a new spark job to use this package

What are notebooks (in Fabric)

- Web-based
- Code first
- Different languages (pyspark, spark, spark SQL and R)
- Native integration with lakehouses

What are Notebooks (in Fabric)

```python
first_parameter = 5
second_parameter = 3
```

✓ 15 sec - Session ready in 14 sec 755 ms. Command executed in 476 ms by jodavid on 10:48:26 PM, 10/27/25

```python
third_parameter = first_parameter * (second_parameter + first_parameter)
```

✓ <1 sec - Command executed in 389 ms by jodavid on 10:48:27 PM, 10/27/25

```python
fourth_parameter= third_parameter - first_parameter
```

✓ <1 sec - Command executed in 348 ms by jodavid on 10:48:27 PM, 10/27/25

```python
display(f"{first_parameter}, {second_parameter}, {third_parameter}, {fourth_parameter}")
```

✓ 2 sec - Command executed in 2 sec 767 ms by jodavid on 10:48:56 PM, 10/27/25

```
'5, 3, 40, 35'
```

Code is in cell based code blocks.
This is easy to debug or test things

What are Dataframes

```python
import pandas as pd

df = pd.DataFrame({
                    'Kingdom':['Gondor', 'Rohan'],
                    'King':['Aragorn', 'Theoden'],
                    'Capital':['Minas Tirith', 'Edoras']
                    })
display(df)
```

	ABC Kingdom	ABC King	ABC Capital
1	Gondor	Aragorn	Minas Tirith
2	Rohan	Theoden	Edoras

A dataframe is a data structure constructed with rows and columns, similar to a database or Excel spreadsheet.

Getting Started

Great, what are we installing

- Semantic link?
- Sempy?
- Semantic link labs?

FOLLOW YOUR NOSE (AKA: CHOOSE)

Microsoft

- Semantic Link
  - Meta-package that depends on the individual parts
- Sempy (or functions-holidays or geopandas or….)
  - Packages that contain core functionality
  - => the individual parts
- Semantic link labs
  - Extends capabilities of Semantic LInk

Community

The Start

- Install/import
- Get data if needed (workspace/item)
- Iterate over list
- Use Semantic link function
- Cleanup
- Write

INSTALL/IMPORT

Semantic Link

- Installed in Fabric by default
- Import it to start

```
import sempy.fabric as fabric
```

Semantic Link Labs

- Needs to be installed

```
%pip install semantic-link-labs
```

- Or create an environment

Environment

Set up shared libraries, Spark compute settings, and resources for notebooks and Spark job definitions.

GET DATA (IF NEEDED)

FROM SL FUNCTION

```
workspaces = fabric.list_workspaces()
sparkdf = spark.createDataFrame(workspaces)
```

FROM LAKEHOUSE

```
Workspaces = spark.sql("""select Id, Name
from LH_SemanticLink_Data.Workspaces""")
```

ITERATING

```
for Id, Name in Workspaces.toLocalIterator():
    print(Name)
```

USE SEMANTIC LINK FUNCTION

Get the list of workspaces

```
Workspaces = spark.sql("""select Id, Name
from LH_SemanticLink_Data.Workspaces""")
```

```
for Id, Name in Workspaces.toLocalIterator():
    temp_items = fabric.list_items(workspace=Id)
```

Iterate over them

By splitting the code, we can work with the list without fetching it each time

CLEANUP

Some columns can’t be saved as received

```
def fnc_PrepareColumns(_Columns):
    _Columns.columns = _Columns.columns.str.replace('[^a-zA-Z0-9]', '', regex=True)
    _Columns.columns = _Columns.columns.str.replace('[ ]', '', regex=True)
    return _Columns
```

THE WRITING

```
workspaces = fabric.list_capacities()
workspaces = fnc_PrepareColumns(workspa
sparkdf = spark.createDataFrame(workspa
sparkdf.write.format("delta").option("m                path}/Tables/{Table_Name}")
```

Explorer

Add lakehouses

- LH_SemanticLink_Data
  - Tables
    - Landing_Fabric_Capacities
    - Landing_Fabric_Columns
    - Landing_Fabric_Items

Overwrite <> append

```
sparkdf.write.format("delta").option("mergeSchema", "true").m   "overwrite"   (f"{lh_abfs_path}/Tables/{Table_Name}")
```

Save to table after creating a dataframe

THE BEACONS ARE LIT

AKA: LIST OF USEFUL THINGS

- LIST_CAPACITIES()
- LIST_WORKSPACES()
- LIST_ITEMS(WORKSPACE)
- LIST_MEASURES(WORKSPACE, DATASET)
- LIST_RELATIONSHIPS(WORKSPACE, DATASET)
- LIST_WORKSPACE_USERS(WORKSPACE)
- LIST_COLUMNS(WORKSPACE, DATASET, EXTENDED)

The Start

This is the logical order

- Install/import
- Get data if needed (workspace/item)
- Iterate over list
- Use Semantic link function
- Cleanup
- Write

But Python can't call anything you haven't declared yet, so the order in the notebooks is different

DeMO

Writing

MODELLING

Star schema all the things

MODELLING

Workspaces

Columns

Users

Semantic Link

Measures

Relations

DeMO

Modelling
& reporting

LAST WORDS OF IMPORTANCE

YOU ARE NOT ALONE

I can't carry it for you... But I can carry you

Allow yourself to be carried

By those who are traveling the same road:

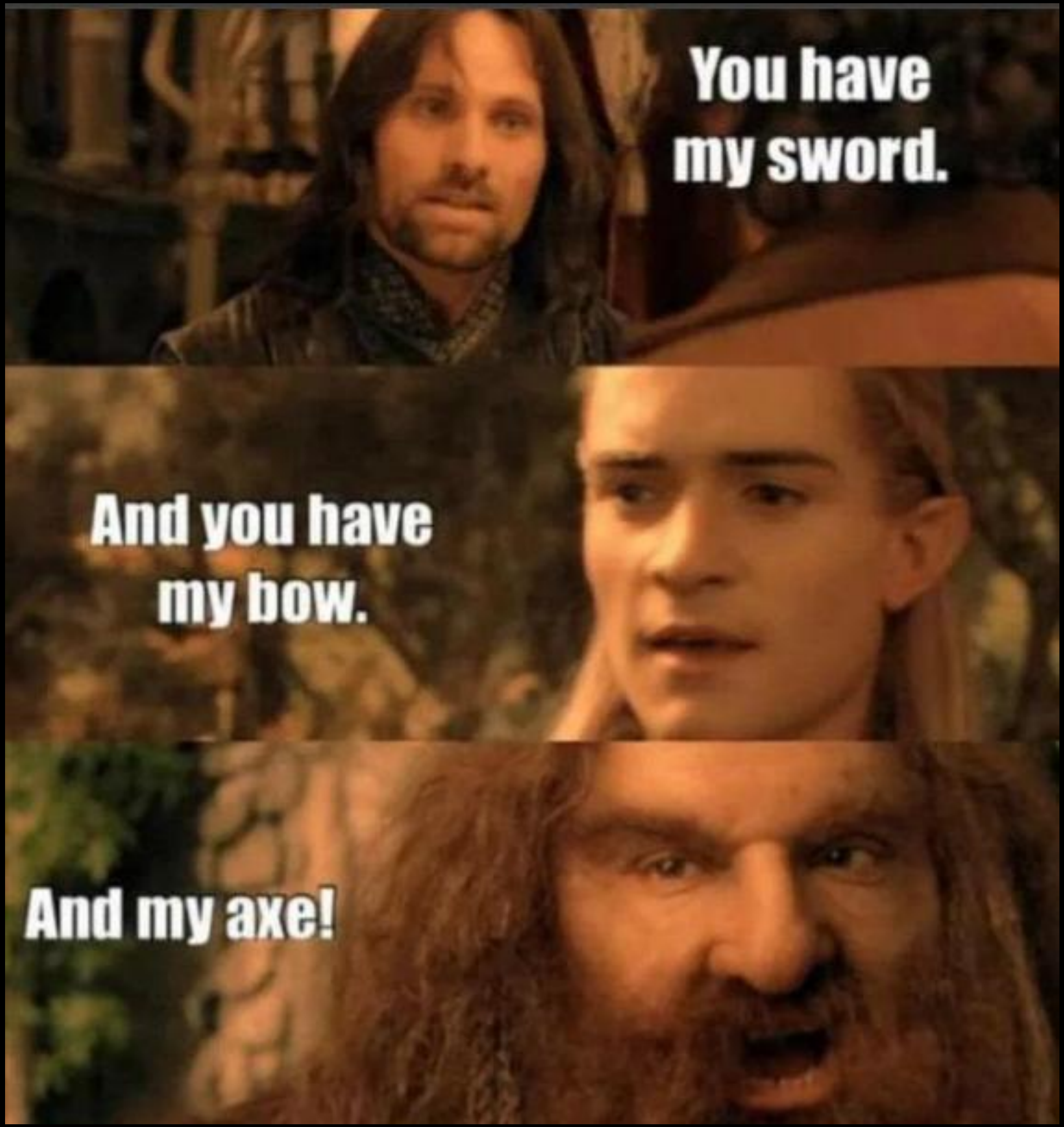
MARC LELIJVELD

DHYANENDRA SINGH
RATHORE-DHYAN

prathy.com

sqlbi.com

fabricguru.com



HTTPS://GITHUB.COM/DATAMINSTREL/DATAMINSTREL_REPO/TREE/MAIN

Thank you for your attention

May semantic link be the light in dark places,
when all other lights go out 

You bow to no one, feedback giver

A wizards introduction to Semantic Link

The reveal

FORODWAITH
RHUN
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Lonely Mountain
Mountains of Mirkwood
Celduin (River Running)
Carnen (Redwater)
MIRKWOOD
East Bight
DOL GULDUR
Sea of Rhûn
Dimrill Dale
LORIEN
Great River
Field of Celebrant
CARDOLAN
Mitheithel
Swanfleet
Tharbad
Baranduin (Brandywine)
MINHIRIATH
Glanduin
Gwathlo (Greyflood)
Eryn Vorn
Lond Daer
ENEDWAITH
Dunland
North-South Road
Isengard
FANGORN
ROHAN
The Wold
The BROWN LANDS
SOUTH UNDEEP
Fords of Isen
Angren (Isen)
Adorn
Helm's Deep
West Emnet
Sarn Gebir
Snowbourn
EASTFOLD
Dead Marshes
Nindalf (Wetwang)
Dagorlad (Battle Plain)
Ash Mountains
Udûn
Isenmouthe
Orodruin
Barad-dûr
Plateau of Gorgoroth
MORDOR
NURN
Sea of Núrnen
EPHEL DUATH (mountains of Shadow)
ERED NIMRAIS
GONDOR
Erech
Lamedon
Pinnath Gelin
Anfalas (Langstrand)
Andrast (Ras Morthil)
Edhellond
Lebennin
Ethir Anduin
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50 100 150 200